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🔗 FR: Configuration Migration Utility

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Draft

Feature request

Component	Complexity	Budget (days)

Server Middle 10

Stakeholders sign off status

Role	Name	Sign-off	Version
Lead product owner	@Aleksandrs Kalimulins	?	1.0
Product owners	@Martins Orinskis	?	1.0
	@Dmitrijs Saldovs	?	1.0
	@Aleksander Vladishev	?	1.0
	@Dmitrijs Jesilevskis	?	1.0
Optional reviewers (none or more)	@Alexei Vladishev	?	1.0
	@Andris Zeila	?	1.0

FR principles

1) focus on user experience first 1.1) in widgets: hide UI elements, instead of disabling them 2) create all UI mockups 3) entity lists must be limited/paginated 4) align with Cloud

Summary

Currently Zabbix does not provide a portable representation of persistent configuration that can be used to move complete configuration state between installations.

Introduce a standalone command-line utility `zabbix_config` for exporting and importing Zabbix configuration independently of the underlying database engine. The utility produces a portable configuration archive containing configuration data serialized as CSV files. The archive is version-specific and can be imported into a clean Zabbix database of the same Zabbix version only.

Use cases

1. As Zabbix administrator I want to migrate configuration from a PostgreSQL-based Zabbix installation to a clean MySQL-based installation.

Zabbix acceptance

1. Create **zabbix_config** utility:
 - a. Standalone command-line utility (zabbix-config)
 - b. Independent of Zabbix Server, Frontend, and Proxy runtime processes
 - c. Uses a database-independent package format
 - d. Reuses existing Zabbix migration and upgrade logic instead of introducing a separate upgrade engine
 - e. Provide machine-readable progress reporting suitable for automation (Cloud-specific req.)
2. **zabbix_config** export:
 - a. parameters:
 - [--db-type mysql|postgresql]
 - [--db-host host]
 - [--db-port port]
 - [--db-name name]
 - [--db-user user]
 - [--db-password password]
 - [--output **bundle_timestamp.tar.gz**]
 - b. On execution:
 - i. connects to the source database
 - ii. exports all database tables to CSV files (Appendix A) except those listed in Appendix B and any user-supplied exclusions
 - iii. preserves original object identifiers and relationships
 - iv. produces **manifest.json** (metadata file) and **checksums.sha256** (SHA-256 digest per file)
 - v. packages all files into **bundle_timestamp.tar.gz** (tar + gzip) following the layout defined in Appendix C
3. **zabbix_config** import:
 - a. parameters:
 - [--input **bundle_timestamp.tar.gz**]
 - [--db-type mysql|postgresql]
 - [--db-host host]
 - [--db-port port]
 - [--db-name name]
 - [--db-user user]
 - [--db-password password]
 - b. On execution:
 - i. disables Foreign key check.

- ii. verifies bundle integrity (checksums, manifest, row counts) before loading any table data
- iii. loads CSV according to the export format (ENCLOSED BY ", ESCAPED BY \, preserving \N as SQL NULL and decoding BLOBs)
- iv. validates post-load row counts per table
- v. recreates ids table entries from imported data after all tables are loaded
- vi. enables Foreign keys check
- vii. automatically upgrades imported configuration to the target Zabbix version reusing the existing Zabbix database upgrade framework

Appendix A — CSV file format

1. First row: quoted header with column names
2. Encoding: utf8mb4
3. Separator: comma
4. Not-NULL fields: always quoted, newlines are allowed inside
5. NULL field: unquoted \N
6. Use backslash escaping for quote and backslash characters
7. BLOB values: exported as Base64 text

Appendix B — Default excluded tables

ids, dbversion, history, history_uint, history_str, history_log, history_text, history_bin, history_json, trends, trends_uint, events, event_symptom, event_tag, event_recovery, event_suppress, problem, problem_tag, alerts, acknowledges, escalations, service_alarms, service_problem, service_problem_tag, item_rtdata, host_rtdata, housekeeper, trigger_queue, lld_macro_export, changelog, sessions, auditlog, task, task_close_problem, task_remote_command, task_remote_command_result, task_data, task_result, task_acknowledge, task_check_now, proxy_history, proxy_dhistory, proxy_rtdata, proxy_group_rtdata, proxy_autoreg_host, dhhosts, dservices, autoreg_host, ha_node

Appendix C — Bundle format

```
bundle_timestamp.tar.gz
├─ manifest.json
├─ checksums.sha256
└─ tables/
    ├─ users.csv
    ├─ hosts.csv
    ├─ items.csv
    └─ ...
```

Archive

- Format: tar + gzip (.tar.gz produced as bundle_timestamp.tar.gz)

- Maximum compressed upload size: **5 GB** (S3 single-PUT limit)

tables/ directory

- One CSV file per exported table
- Filename format: <table_name>.csv

manifest.json

Defines general attributes and basic metadata per table. Schema:

```
{
  "version": "1.0",
  "zabbix_version": "7.0.5",
  "dbversion_mandatory": 7000000,
  "created_at": "2026-07-01T10:00:00Z",
  "table_count": 161,
  "tables": [
    { "table": "role", "file": "tables/role.csv", "rows": 4 },
    { "table": "hosts", "file": "tables/hosts.csv", "rows": 312 }
    ...
  ]
}
```

checksums.sha256

- One line per file in the bundle, covering all three metadata files and all CSV files
- Format: standard sha256sum output — <hex_digest> <relative_path> (two spaces between digest and path), one entry per line, paths relative to the archive root
- Example

```
e3b0c44298fc1c149afb... manifest.json
a87ff679a2f3e71d9181... tables/users.csv
d41d8cd98f00b204e980... tables/role.csv
```

Zabbix UI changes

N/A

Open questions

N/A

Decisions made

1. Decision

Change log

N/A



0 Likes

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Tags:

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